

23 Which rate equation is correct for the reaction between X, Y and Z?

| experiment | initial concentrations / mol dm <sup>-3</sup> |      |      | rate<br>/ mol dm <sup>-3</sup> s <sup>-1</sup> |
|------------|-----------------------------------------------|------|------|------------------------------------------------|
|            | X                                             | Y    | Z    |                                                |
| 1          | 0.01                                          | 0.10 | 0.01 | $2.00 \times 10^{-4}$                          |
| 2          | 0.01                                          | 0.10 | 0.02 | $4.00 \times 10^{-4}$                          |
| 3          | 0.02                                          | 0.10 | 0.01 | $4.00 \times 10^{-4}$                          |
| 4          | 0.02                                          | 0.20 | 0.01 | $4.00 \times 10^{-4}$                          |

- A rate =  $k[X][Y]$   
 B rate =  $k[X][Y]^2$   
 C rate =  $k[X][Y][Z]$   
 D rate =  $k[X][Z]$